

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 3 – Viva Based Questions

|                  |                                                                                                                            |               |                                          |
|------------------|----------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------|
| Course Code:     | DSA0404                                                                                                                    | Course Title: | FUNDAMENTALS OF DATA SCIENCE             |
| CO Assessed:     | CO4 – Viva Based Questions                                                                                                 | Assessment:   | Assessment Tool 3 – Viva Based Questions |
| Weightage:       | 20% of CIA                                                                                                                 | Total Marks:  | 100                                      |
| CO4 Statement:   | Excellent conceptual understanding, confident explanation, accurate interpretation, and strong industry application. (BL4) |               |                                          |
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| Section / Batch: | B.E-CSE-AI                                                                                                                 | Date:         | 21-08-2026                               |

### Code of Conduct

I Shaik Nuha(192472242) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sk. Nuha

| Criteria                | Weight age | Level 4 – Exemplary                                              | Level 3 – Proficient                           | Level 2 – Developing                    | Level 1 – Beginning                                  | Marks |
|-------------------------|------------|------------------------------------------------------------------|------------------------------------------------|-----------------------------------------|------------------------------------------------------|-------|
| Understanding of Case   | 25         | Thorough understanding; clearly identifies all key issues        | Good understanding with most issues identified | Basic understanding; some issues missed | Poor understanding; problem not identified correctly |       |
| Application of Concepts | 25         | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps   | Limited application of concepts         | Fails to apply relevant concepts                     |       |
| Analysis                | 20         | In-depth analysis with strong reasoning and insights             | Good analysis with logical reasoning           | Basic analysis with limited depth       | Weak or no analysis; lacks reasoning                 |       |
| Solution Design         | 20         | Innovative, feasible solutions with strong justification         | Logical solutions with adequate justification  | Basic solutions with weak justification | Incorrect or no solution provided                    |       |
| Clarity                 | 10         | Clear, well-structured, and easy to understand                   | Organized and understandable presentation      | Limited clarity and structure           | Poor clarity; difficult to understand                |       |

## **A. Introduction to Statistical Inference**

### **1. What is statistical inference?**

Statistical inference is the process of using sample data to draw conclusions or make predictions about a population.

### **2. Why is statistical inference important in data science?**

It helps data scientists make decisions about a large population using limited sample data and quantify the uncertainty in those decisions.

### **3. What is the difference between descriptive statistics and inferential statistics?**

- **Descriptive statistics:** Summarizes the data we already have.
- **Inferential statistics:** Uses sample data to make conclusions about a larger population.

**Example:** Average marks of 50 students are descriptive; using those 50 students to estimate the average marks of all students is inferential.

### **4. What is a population in statistics?**

A population is the complete group of individuals or observations that we are interested in studying.

**Example:** All students in a university.

### **5. What is a sample?**

A sample is a smaller subset selected from the population for analysis.

**Example:** 200 students selected from the university.

### **6. Why do data scientists usually work with samples instead of complete populations?**

Because studying the entire population can be expensive, time-consuming, or sometimes impossible.

### **7. What is a population parameter?**

A population parameter is a numerical value that describes a population.

Examples:

- Population mean  $\mu$
- Population proportion  $P$

- Population variance  $\sigma^2$

### 8. What is a sample statistic?

A sample statistic is a numerical value calculated from sample data.

Examples:

- Sample mean  $\bar{x}$
- Sample proportion  $\hat{p}$
- Sample variance  $s^2$

### 9. Give a real-world example where statistical inference is used in industry.

An e-commerce company can survey a sample of customers to estimate the satisfaction level of all its customers.

### 10. What are the two major activities involved in statistical inference?

The two major activities are:

1. Estimation
2. Hypothesis testing

## B. Frequentist Approach

### 11. What is the frequentist approach to statistical inference?

It is an approach where probability is interpreted as the long-run frequency of events, and population parameters are treated as fixed values.

### 12. What is the basic idea behind frequentist statistics?

The basic idea is to use repeated-sampling behaviour of data to estimate parameters and test hypotheses.

### 13. How does the frequentist approach differ from the Bayesian approach?

| Frequentist                            | Bayesian                                            |
|----------------------------------------|-----------------------------------------------------|
| Parameter is fixed                     | Parameter is treated as uncertain/random            |
| Uses sample data                       | Uses data + prior information                       |
| Uses p-values and confidence intervals | Uses posterior probabilities and credible intervals |

**14. In the frequentist approach, what is considered fixed: the parameter or the data?**

The population parameter is considered fixed, while the sample data is considered random.

**15. What does long-run frequency mean in frequentist statistics?**

It means the proportion of times an event occurs when an experiment is repeated a very large number of times.

**16. Why is random sampling important in the frequentist approach?**

Random sampling helps ensure that the sample is representative of the population and allows valid statistical inference.

**17. Can a population parameter be considered a random variable in the frequentist approach? Why?**

No. In frequentist statistics, the population parameter is considered a fixed but unknown value, not a random variable.

**18. Give an example of a frequentist problem in a data science application.**

A company may test whether a new recommendation algorithm increases the conversion rate compared with the existing algorithm using a hypothesis test and p-value.

## **C. Point Estimation**

**19. What is a point estimate?**

A point estimate is a single numerical value used to estimate an unknown population parameter.

**Example:** Using a sample mean of 75 to estimate the population mean.

**20. What is an estimator?**

An estimator is a statistical rule or formula used to estimate a population parameter.

**21. What is the difference between an estimator and an estimate?**

- **Estimator:** The formula or method.
- **Estimate:** The numerical result obtained from that formula.

Example:

$$\bar{x} = \frac{\sum x}{n}$$

is the estimator, while 78 is the estimate.

**22. What is the point estimator for the population mean?**

The sample mean:

$$\pi = \frac{\sum x_i}{n}$$

**23. How do you estimate a population proportion from a sample?**

Use the sample proportion:

$$\hat{p} = x/n$$

where (x) is the number of successes and (n) is the sample size.

**24. What is the point estimate of population variance?**

The sample variance:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

**25. Why is the sample mean commonly used to estimate the population mean?**

Because the sample mean is unbiased, consistent, and efficient under common assumptions.

**26. What properties should a good estimator have?**

A good estimator should ideally be:

1. Unbiased
2. Consistent
3. Efficient

**27. What is an unbiased estimator?**

An estimator is unbiased if its expected value equals the true population parameter.

$$E(\hat{\theta}) = \theta$$

**28. What is meant by the bias of an estimator?**

Bias is the difference between the expected value of an estimator and the true parameter.

$$Bias(\widehat{\theta}) = E(\widehat{\theta}) - \theta$$

**29. What is the difference between bias and variance of an estimator?**

- **Bias:** Measures how far the estimator is systematically from the true value.
- **Variance:** Measures how much the estimator changes from sample to sample.

**30. Why can two different samples produce different point estimates for the same population parameter?**

Because different random samples contain different observations, causing the calculated statistics to vary.

**D. Variability in Estimates**

**31. Why does a point estimate have uncertainty?**

Because it is calculated from a sample rather than the entire population, and different samples can give different results.

**32. What is the sampling distribution of an estimator?**

It is the probability distribution of an estimator obtained from repeated samples of the same population.

**33. What is the standard error?**

Standard error measures the variability of a sample statistic or estimator across repeated samples.

For the sample mean:  $SE = \sigma / \sqrt{n}$

**34. How is standard error different from standard deviation?**

- **Standard deviation:** Measures variability among individual observations.
- **Standard error:** Measures variability of a sample statistic across samples.

**35. How does increasing the sample size affect the standard error of the sample mean?**

$$SE = \frac{\sigma}{\sqrt{n}}$$

As (n) increases, the standard error decreases.

**36. Why does variability in an estimate decrease when the sample size increases?**

Larger samples contain more information about the population, so random sampling fluctuations become smaller.

### **37. What is the relationship between standard error and confidence intervals?**

The standard error determines how much uncertainty is around the estimate.

Generally:

$$CI = \text{Estimate} \pm \text{Critical value} \times SE$$

A larger SE produces a wider confidence interval.

### **38. If two samples have different sample means, does it necessarily mean that one sample is incorrect?**

No. Different random samples naturally produce different sample means. This is called sampling variability.

## **E. Confidence Intervals**

### **39. What is a confidence interval?**

A confidence interval is a range of plausible values for an unknown population parameter, calculated from sample data.

### **40. Why is a confidence interval more informative than a point estimate?**

A point estimate gives only one value, while a confidence interval gives a range and indicates the uncertainty around the estimate.

### **41. What does a 95% confidence level mean in the frequentist approach?**

If we repeatedly take samples and construct 95% confidence intervals using the same method, approximately 95% of those intervals would contain the true population parameter.

**Important:** It does not strictly mean there is a 95% probability that the particular calculated interval contains the parameter.

### **42. What is the difference between a confidence level and a confidence interval?**

- **Confidence level:** The percentage such as 90%, 95%, or 99% that describes the reliability of the interval-building procedure.
- **Confidence interval:** The actual numerical range calculated from the sample.

### **43. What factors determine the width of a confidence interval?**

Main factors are:

1. Sample size
2. Data variability
3. Confidence level

Higher variability and higher confidence level make the interval wider, while larger sample size makes it narrower.

### **44. How does increasing the sample size affect the width of a confidence interval?**

Increasing the sample size generally makes the confidence interval narrower because the standard error decreases.

### **45. How does increasing the confidence level from 95% to 99% affect the confidence interval?**

The interval becomes wider, because we need a larger range to achieve higher confidence.

### **46. What is the difference between a z-confidence interval and a t-confidence interval?**

#### **Z-confidence interval:**

- Uses the normal distribution.
- Commonly used when population standard deviation is known or sample size is sufficiently large.

#### **T-confidence interval:**

- Uses the t-distribution.
- Commonly used when population standard deviation is unknown, especially for smaller samples.

### **47. When would you use the t-distribution instead of the normal distribution for estimating a population mean?**

When the population standard deviation is unknown, particularly when the sample size is small.



## F. Hypothesis Testing Using Confidence Intervals

### 48. What is hypothesis testing?

Hypothesis testing is a statistical method used to determine whether there is enough evidence to support or reject a claim about a population.

### 49. What are the null hypothesis ( $H_0$ ) and alternative hypothesis ( $H_1$ )?

**Null hypothesis ( $H_0$ ):** Represents the default claim or no significant effect.

**Alternative hypothesis ( $H_1$ ):** Represents the claim that we want to find evidence for.

**Example:**

$$H_0: \mu = 50$$

$$H_1: \mu \neq 50$$

### 50. How can a confidence interval be used to make a decision about a hypothesis?

Compare the claimed parameter value with the confidence interval.

For a two-sided test at 5% significance:

- If the claimed value is inside the 95% CI  $\rightarrow$  generally fail to reject  $H_0$ .
- If the claimed value is outside the 95% CI  $\rightarrow$  generally reject  $H_0$ .